

William Li

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Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, C++, SQL, Bash, HTML, CSS

Frameworks: React, Next.js, Express, Node.js, React Native, REST API

Technologies: Git, Docker, Roblox Studio, Linux (Ubuntu), PostgreSQL, MySQL, MongoDB, SQLite

Experience

Software Engineer Intern

Jun 2026 – Aug 2026

Vici Learning

Hybrid

- Built an admin-facing integration dashboard that automatically syncs booking data from **SimplyBook.me** into a local **PostgreSQL** database, eliminating the client's weekly manual Excel reconciliation process.
- Implemented a **rules engine** to surface actionable tasks (lapsed students, low upcoming booking volume, cancellation spikes, and unpaid/unbooked flags) with configurable alert thresholds.
- Integrated **Brevo** (CRM/email) to automate transactional communications (session reminders, payment reminders, membership renewals, lapsed-student follow-ups), with a human-approval review queue for personal messages.
- Designed a sync architecture using **Spring @Scheduled** background jobs and a custom **JSON-RPC 2.0** client, making the dashboard immune to API rate limits and upstream downtime.
- Delivered the full system as a one-command **Docker Compose** package with an admin user manual, operations runbook, and a recorded handoff training session for non-technical staff.

Projects

Escape From Windows | React, TypeScript, Node.js, Express, Docker, REST API, Java, Python

Jan 2026

- 🏆 **SystemHacks 2026: Most Outstanding Project (First Memory Byte Award)**
- Built a full-stack browser-based puzzle game using **React**, **TypeScript**, and a **Node.js/Express REST API** to manage persistent game state, authentication, and session data.
- Developed a **Python-based AI hint engine** that analyzes player interaction patterns and dynamically generates contextual hints using heuristic modeling.
- Containerized the application with **Docker** and **Docker Compose**, ensuring reproducible development and production environments during a 24-hour hackathon.
- Implemented automated CI/CD pipelines with **GitHub Actions**; deployed frontend to **Vercel** and backend services to **Render**.
- Designed a simulated OS-style UI with modular **React component architecture**, enabling real-time puzzle triggers via structured **client-server API communication**.

Office Hours Queue & Analytics Platform | React, Node.js, Express, PostgreSQL, WebSocket, Google Calendar API

June 2026

- Collaborated on a 5-person team to design and build a real-time office hours queueing platform for high-demand university courses, replacing ad hoc Google Docs and Zoom waiting rooms with a self-serve queue.
- Owned authentication and authorization, implementing session-based auth and **Role-Based Access Control (RBAC)** middleware to separate Student, TA, and Instructor views and permissions.
- Built secure session management and access-control logic supporting real-time **WebSocket** state shared across the web client.
- Contributed to a **Node.js/Express REST API** on **PostgreSQL**, deployed to a Google Cloud VM behind an **Nginx** reverse proxy, and integrated with the **Google Calendar API** for session scheduling and notifications.

Education

Simon Fraser University

Sept 2024 – Present

Bachelor of Science in Software Systems

Burnaby, BC

Relevant Coursework

Data Structures and Programming, Object-Oriented Design (Java), Introduction to Software Engineering, Probability and Computing, Introduction to Computer Systems, Client-Side Development, Probability and Statistics for Computer Science

Leadership & Teaching Experience

STEM Tutor

May 2023 – Jun 2024

RoboKids

Remote

- Delivered instruction in **Scratch**, **Python**, **Java**, robotics, and 3D design to young students.
- Guided students through **computational thinking** and project-based learning methodologies.